

The Science of Resonances, in the Intimate Nature of the Universe.

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Date: April 10, 2024

Last modified: 18 July 2024

Summary:

In the scientific contemporary history, the "determinism" ensures that we perceive with our senses the infinite possibilities of being that atoms allow to form, since we have the culture of atomism, which we are overcoming in the name of dynamic holism. In the dynamic holism, mathematics and geometry are the means that we use to represent reality, but these are not sufficient if they are not accompanied by our "sixth sense", also called intuition; it follows that any research process that is not accompanied by an intellectual passion is destined to fail. From this perspective, Science must admit that the emotional field is part of the scientific knowledge and with this admission, it will not only save itself, but through its example it will support the entire system of cultural and emotional life of which it is an integral part.

The "Science of Resonances", which comes from our passion for understanding what we are, and what we are doing on this immense and harmonious "Brilliant Garland" called the Universe, accompanies us in the search for the "reasons", at the basis of both various crises that humanity is going through, and of the mechanisms that regulate the Universe. These mechanisms start from the concept of "wave-particle" dualism, which in the "Science of Resonances" is replaced with that of "Field and Resonance". In the "Field Resonance" there is no separation between the "condensed matter" and the "Quantum Void" or "Space" or "Ether", from where everything is born and everything returns enriched by the experience gained in the passage into the state of imperfection. The state of imperfection in the Eastern philosophies, as for example in the Chinese Tao, it is seen as the vital spark (Qi) that arises from the accumulation of the two opposites Yin and Yang, while in the Ayurvedic conception, the being resonates on harmonious frequencies in a continuous evolution, which go beyond the passage in life in the so-called condensed matter.

The "Science of Resonances" also teaches us that the "Universal Evolutionary Arrow" is generated by a pulsation in the shape of a Fibonacci Golden Spiral, which oscillates between two limits, of which the lower limit never passes from the value 0, but passes from value 0.5, which represents the state of minimum energy possible in the Quantum Vacuum or Space, and the upper limit, is seen as the interruption of the evolution of a perfect evolutionary process, well represented by the Riemann hypothesis.

The Universal pulsation, therefore, turns out to be the first generating act that forms Time which, in turn, generates the first condensed mass, the hydrogen atom. The latter, through analogue and non-digital logic, has the possibility of making its electron "jump" across seven energy levels, which create a non-chaotic "energy symphony" capable of respecting the functioning of Nature, at opposite of the probabilistic representation given to us by the Schrödinger equation. In this analogical conception of Nature, the concepts of Intelligence and Consciousness are not separable, and have always represented the great questions of those who decide to deal with the mysteries still inherent to the Universe, and where to justify the magic of Universal Life, the presence of an intrinsic Universal Intelligence.

Finally, it is interesting to note that the "Science of Resonances", applied in Technology, will lead to important spin-offs in the fields of nano-technologies, new composite materials, health in Biological Life, means of Information Transmission, and means of Transport.

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1) Introduction

By looking and listening to the Universe carefully, we would realize that it was as if for millennia we had observed the sky only in "black and white", regardless of how much energy and/or information reached us in the form of frequencies and/or lengths of wave not perceptible to us using only our senses and the equipment of our current technology. In fact the radiation visible with our eyes represents only a very small part compared to the entire electromagnetic spectrum, which embraces wavelengths that are even millions of times shorter than those of the visible, as in the case of very short gamma radiation, up to hundreds of millions of times longer as in the case of long radio waves. This is in addition to the fact that, probably, in addition to the spectrum of electromagnetic waves already known to us, in living and conscious Space, there is still something else that we have not yet discovered. Radio astronomers were the first to begin investigating the Universe in the form of frequencies, and therefore no longer just in an optical-visual form; classifying waves based on their energy, frequency, as well as the amount of information they carry.

In this article we want to go beyond the work already carried out by radio astronomers and astrophysicists, addressing in their profound essence the various **"whys"** of the harmonic relationships that regulate the frequencies, which, in our common vocabulary are cited as: particles, fermions, bosons, strings, super-strings, harmonic strings, energy strings, computer strings, stable ether, unstable ether etc. with many different names to express "essences" equal in substance, but different in forms and/or functions, which go to compose that enormous and magnificent "Harmonic Mosaic" in continuous evolutionary transformation which is the Universe, where in its various manifestations of existence, present both in the condition of non-tensioned Ether, or more simply Space, and Tensioned Ether, or condensed matter; it is full of sensations that we, with our humble senses, perceive only in a small part. The wise man says: *"Looking out the window, we thought we were discovering a new Universe. It took us years to understand that there was a cat on the windowsill, and the new Universe was us reflected upside down in its eye"*. This short quote wants us to understand that many times, what we are looking for, has always been in front of us, we just need to find the right attention and sensitivity to make it usable in our conscious Consciousness; but our usual problem is that we observe little, and we listen even less.

2) In the beginning was Perfection, birth of the illusion of Time, and Function of the Fibonacci Golden Spiral Pulsation

In the beginning it was Perfection, from which a state of imperfection emerged, which through various life experiences tends to return towards Perfection, see: **Ivaldi, Smigliani, 2024a**; where reality manifests itself in an apparent *"pulsating Chaos"*, which we interpret as Time, in which Universal events do not all sing the same score in chorus; from here it arises the dilemma of not being able to explain the natural evolutionary order, which we understand to be intrinsic to the global system. In this way, by giving up predicting events precisely, in order to verify the chances of success of our predictions, we accept the concept of *"Indeterminacy"*, thus basing a scientific explanation on a probabilistic logic; from this dilemma *"quantum mechanics"* was born.

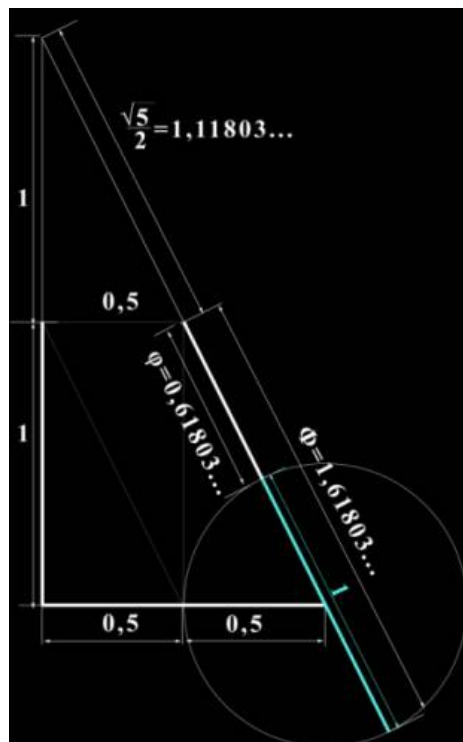
In contemporary science, the problem of the relationship between Space and Time is still the subject of open debate, where for Aristotle Time represented the way of measuring how things move, seen as a linear sequence of succession of events, and/or as a sequence of the motion of a photon in space, perceived by us as absolute. In reality with our clocks and our instruments, we cannot measure absolute time, but we can only measure the frequency, the speed, and the sequence inherent in the state of change of condensed matter.

To build his physics, Newton imagined an empty space where time passes, even if there is nothing and nothing happens. In this context, time represents the dimension in which the passing of events is conceived and measured, which produces the distinction between past, present and future.

In the last century, Einstein's Relativity profoundly modified this conception of Space and Time, where Einstein considers Space as the container of reality, excluding the existence of the ether. In this context, through the constant "c" of the speed of light, Time is no longer seen as the flow of existence, but Space and Time become "Relative" entities, within a four-dimensional "continuum", which by doing without the presence of the Ether, justifies an empty Space, which does not correspond to Reality. In fact Einstein in 1924 wrote: *"In theoretical physics, we go nowhere without the ether, that is, a continuum of physical properties defined, since the general theory of relativity excludes long-range direct action; and every short-range theory assumes the presence of continuous fields, and, consequently, the existence of the ether."*

In the "Science of Resonances" the entities manifest themselves through a generating wave which, in order to "self-sustain" without being "self-destructive", shares the Space through the shape of the Fibonacci golden spiral, which has limits both in the low and high of its unfolding: the lower part has the characteristic of never passing from 0, but from 0.5, is understood as the state of minimum possible energy in the Universe, while the limit of the upper part is well represented by the Zeta function of Riemann, see: **Smigliani, Ivaldi, 2024.**

Engineer Francesco De Santis with his book "Algorithms and geometries of prime numbers", see: **De Santis 2019**, had made a good contribution to the deepening of this knowledge, in which as illustrated on Page 13 of Chapter 6) *Elegance Harmonic in Mathematics and Geometry*, Engineer De Santis explains to us that to understand the functioning of these mechanisms, we must first clarify the meanings attributed to the term Golden Ratio (Golden Ratio or Golden Section), which is the basis of everything that moves in the Universe, where the problem would be solved if the large Phi (in English Capital PHI) was the largest value, i.e. 1.618033, and the small phi (in English Small letter phi), and here we wanted to assign vulgar names" Small letter", phi would also historically be 0.618033..., and Capital PHI 1.618033... could finally be called "Golden number", which in truth was the number 235 of Meton's cycle, engraved in letters of gold on the columns of the temple of Athena in Greece: 2; 3 and 5 which are the numbers of the Fibonacci series formed by the construction triplet 0;1;1.



Fibonacci Succession: the Metone column contains the Golden Number written in Gold letters on the columns of the temple of Athena. 235 moons correspond to 19 solar years.

	Metone			Fibo	Lucas		
	0			0		0	
13		3		1	1	5	
	13		3	1			25
48		10		2	4	20	
	35		7	3		15	75
125		25		5	11	55	
	90		18	8		40	200
325		65		13	29	145	
	235		47	21		105	525
850		170		34	76	380	
	615		123	55		275	1375
2225		445		89	199	995	
	1610		322	144		720	3600
5825		1165		233	521	2605	
	4215		843	377		1885	9425
15250		3050		610	1364	6820	
	11035		2207	987		4935	24675
39925		7985		1597	3571	17855	
	28890		5778	2584		12920	64600
104525		20905		4181	9349	46745	
	75635		15127	6765		33825	
		54730		10946	24476		
			39603	17711			
				28657			

Figure 1. Pythagoras' secret on the left, and on the right the numbers of the sequences connected to that of Fibonacci

Engineer De Santis, see: **Figure 1**, teaches us that the problem comes from very far away, from Pythagoras' religious secret of not wanting to reveal the irrationality of the root of 5 divided by 2 and which is 1.118033. where adding and subtracting 0.5 gives us our two numbers. Pythagoras, (the guru of the penta would sound in Calabrian dialect: u pintaguru and by elision of the n u pitaguru the Pythagoras) of the pentagon, in the shape of a 5-pointed star, had made the emblem of his school in Crotone, and knew the fact that the equilateral triangle with sides 1 and 2 leads to the root of 5 etc.

3) From Greek thought to Quantum physics, and the “Collapse of the Wave Function” crisis.

We are children of Greek thought, which is used with Plato on the one hand, and Democritus on the other, seeking the future Truth, where in our passage into imperfection, see: **Ivaldi, Smigliani, 2024a**, we must know how to cultivate with care and dedication “*doubt*”, as it represents the stimulus that pushes us towards knowledge and awareness of what we are in reality. In Plato's *Timaeus* the geometric vision of the Cosmos is based on the harmony of the golden section and on the solids that Plato considered all lying in the primordial form of the sphere: tetrahedron, octahedron, cube, and icosahedron, with equal faces, vertices and angles. They are connected to each other in correspondence with air, water, earth, fire, and ether (seen as a “fifth essence”), and represent a milestone in the era of Platonic thought: the thesis supported by Plato says, in fact, that the tiny “*particles*” that make up matter correspond to mathematical figures which in reality are not matter, but geometric representations of “*symmetry of Space*” groups.

The Platonic theory, due to its random and not very intuitive character, was less successful than the atomistic theory of Democritus, even if we find Plato's thought present in recent scientific history, in the thought of W. K. Heisenberg (1901-1976), which captures in the anti-intuitivity of Platonic theory, a profound assonance with the quantum description of reality, accused of illogicality and inconsistency precisely because of its distance from our "*materialistic logic*".

With Werner Karl Heisenberg, quantum mechanics begins to support the idea that today's physics must be substantially based on the hypothesis that the fundamental laws of nature are symmetrical, and invariant, with respect to the use of certain symmetry groups, even if they are symmetry groups different from those dealt with by Plato.

The problem of the ultimate consistency of reality, and of the dissolution of the concept of matter, was addressed by Erwin Schrödinger (1887-1961), who as a physicist challenged the corpuscular interpretation of reality, in favor of the wave one, and as a philosopher he converted his vision of the "*continuum of reality*", in a reference to Plato, and to eastern philosophy.

It should be remembered that a good contribution to this scientific vision was given by Max Planck (1858-1947), who at the beginning of the 20th century introduced the concept of "*quantum*", as the basic constituent element of energy and mass.

Many scientists have asked themselves the question of how Nature proceeds in its evolutionary choices, starting with Albert Einstein who pronounced the famous phrase: "*God does not play dice with the Universe*", which was addressed to Niels Bohr in a letter in where they discussed the interpretation to be given to quantum mechanics. What Einstein could not accept was the random character of quantum theory, which was revolutionary for the deterministic thinking derived from centuries of classical physics; Bohr's response was: "*Einstein, don't tell God what to do.*"

The question posed by Quantum theory, which investigates the way in which objective reality emerges from the palette of possibilities provided by nature, is still the subject of century-old arguments; recent experiments, performed with much more sensitive instruments than the past, to test the idea whether Erwin Schrödinger's "collapse of the wave function" was not just the result of "*mathematical convenience*" but, instead, a real physical process, they are not finding good evidence, see: **Stoica 2017**.

Here the big question arises: what causes spontaneous localization via wave function collapse?

The models don't say; but they suggest adding a mathematical term to the Schrödinger equation to describe it; while for the "Science of Resonances", see: **Ivaldi, Smigliani, 2023**, the Universe is life which also manifests itself in biological evolution, which depends on a set of primordial rules having good links with the pilot wave theory of **Bohm, 2006**, where the concept of field is seen as an implied order, from which it can be deduced that "*Quantum*", deriving from statistical science, is not suited to universal reality, as it does not have within it the tools to represent the birth and evolution of Universal Life. This representation requires instead the acceptance of the presence of an intrinsic Universal Intelligence, see: **Meijer, Ivaldi, 2022**, much more suitable for representing the "*Evolutionary Arrow*" inherent in Universal Reality, the mechanism of which can be seen further Fibonacci, and beyond Riemann, see: **Ivaldi, Smigliani, 2024b**.

4) From the Fine Structure Constant to Universal Geometry

In Classical Physics the energy levels of the hydrogen atom are established starting from the atom model postulated by Bohr, where the electrons move around the nucleus on circular orbits, or as later suggested by Arnold Sommerfeld (1868-1951), on elliptical orbits, each with a certain "allowed energy", and with "relativistic dependence" of the mass on the velocity, which leads to a number indicated with the Greek letter α , whose numerical value is close to the ratio $1/137$, where to explain it in Classical Physics, it is necessary to characterize the intensity of the electromagnetic force acting on charged particles such as electrons and protons.

The fact that its value, $1/137$, is small means that electromagnetism is a relatively weak force, and consequently that electrons are able to "jump out" of atoms and form chemical bonds. We note here that if it were even smaller, say $1/138$, stars would not have been able to form carbon atoms, and biological life would not have been able to exist, at least as we know it.

A fundamental point of reference in the Laws of Classical Physics is represented by the "*Universal Constants*", of which one of the most important is the "*Fine Structure Constant*", designed to measure the intensity of the electromagnetic field relating to the elementary charges that "*couple*" interacting with each other, like the electron with the photon, practically the amplitude with which a real electron emits or absorbs a real photon, and is dimensionless because it is related to two other fundamental constants: $\hbar$ and c (for the system of units electrostatic measurement CGS where K , Coulomb constant, has value 1).

$$\alpha = \frac{e^2}{\hbar c}$$

e = charge of the electron, c = speed of light, with a value equal to

$$\frac{1}{137} = 0,007299$$

The inverse square of this sum corresponds to:

$$\alpha = \frac{e^2}{q_p^2} = \frac{1}{11,706} = 0,08542455$$

where

$$\alpha = \frac{e^2}{q_p^2}$$

is the ratio of the electron energy to the Planck elementary charge.

For classical science, the constant of the fine structure is a dimensionless number (like Planck's constant), which if analyzed with purely mathematical expressions alone, it is difficult to grasp its profound meaning. While for the "Science of Resonances", see: **Ivaldi, Smigliani, 2023**, the fine structure constant is linked to the model of a pulsating vortex in the shape of the Fibonacci Golden Spiral, see: **Pietro, 2006**, in a Quantum Void with the characteristics of a neural superfluid, see: **Scala 2022a**.

This coupling number, which for classical science is still an unrevealed mystery to understand where it comes from, for the "Science of Resonances" can be seen as the existing relationship between octahedron and tetrahedron, when energy moves by pulsing from one to another, and you can also see it in the residual energy that is produced when you collapse a sphere into a cube, or expand a cube into a sphere, where the volume of the two objects is equal, the only difference between cube and sphere, lies in the total surface area, in which the surplus surface between the two is equivalent to a series of unsuspected constants, including that of the fine structure, so we ask ourselves: what should be the relationship between the radius of said sphere and the edge of the cube in question? Let's try to figure it out:

$$\frac{4}{3}\pi r^3 = L^3$$

So:

$$r^3 = L^3 \frac{3}{4\pi}$$

From which:

$$r = L \sqrt[3]{\frac{3}{4\pi}}$$

So:

$$r = L \sqrt[3]{\frac{3}{12,56}}$$

Therefore:

$$r = L \sqrt[3]{0,238}$$

With final result:

$$r = 0,619L$$

From which it can be seen that it corresponds to the golden ratio par excellence (0.6189.....).

If we then consider the sphere and the cube, according to the "Science of Resonances", as the energy of Space condensed into mass, we could consider the value of the edge of the cube as a fundamental energy unit equal to **0.5**, therefore:

$$r = 0,619 \cdot 0,5 = 1,119$$

Meaning what:

$$\frac{\sqrt{5}}{2}$$

Of Pythagorean memory.

In addition, the gold number 0.619 is divisible by 0.085, the result of the root of the ratio of electron charge, e , to Planck's minimum elementary charge, q_p

$$\alpha = \frac{e^2}{q_p^2}$$

representing the inverse of the square of the fine structure with value

$$\alpha = \frac{e^2}{h \cdot \text{tagl} \cdot c} = \frac{1}{137} = 0,007299$$

The result of division between 0.619 and 0.085 is given not surprisingly by 0.00729×103 .

If we then derive the areas of the two solids with even volumes, it will be found that:

$$A_{SF} = 4\pi r^2$$

A_{SF} = area of the sphere, and:

$$A_c = 6L^2$$

A_c = area of the cube.

If we set $L = 0.5$ (minimum energy value in Space, with frequency **0.5** Hz and pulsation 3.14), then $r = 0.619 \times 0.5 = 0.309$, so

$$A_{SF} = 4\pi \cdot 0,309^2 = 0,095$$

$$A_c = 6 \cdot 0,5^2 = 1,5$$

The area of the cube exceeds that of the sphere by 1,405, which tells us that the static component (cube) of the Universe's basic energy (**0.5**) is far greater than the dynamic component (sphere, which accounts for 6.3 percent relative to the cube).

If we set $L=3.14$ (minimum energy pulsation of the Universe, **0.5**), imagining that a pulsation/information with harmonic characteristics is triggered on the static surface (cube), we will have that $r=0.619 \times 3.14=1.943$, so:

$$A_{SF} = 4\pi \cdot 1,943^2 = 47,417$$

$$A_c = 6 \cdot 3,14^2 = 59,157$$

In this case, the dynamical surface is very close to the static surface (about 80% of it), as if the triggering of the pulsation created much more "*dynamic*" possibilities (including life).

The difference in the areas of the two solids with equal volume (cube and sphere), therefore, for the "Science of Resonances" represents the pulsation derived from the mass condensation of the pulsating information of the Universe, which is conditioned by the fine structure constant, which in turn, as we have seen before, is correlated with the golden number and with the minimum energy of Space, both as a frequency (equal to **0.5Hz**) and as a pulsation (π), see: **Ivaldi, Smigliani, 2024a**. Then again, $\pi^2 = 9.860... \approx g$; will this also be a **case**?

At this point we ask ourselves how can the constant of the fine structure be a ratio between octahedron and tetrahedron, and at the same time also be a ratio between the cube and the sphere?

For the "Science of Resonances", this is an aspect of symmetry in action, from which it can be deduced that different geometric shapes can have similar properties, since they are all incorporated into each other with perfectly harmonious relationships, demonstrating to us that they exist geometrically structured pulsating etheric energies at work in the atom.

This scientific vision was already present in the ancient philosophers, who believed that the Universe was full of what they called the "*Cosmic Network*", in which each cell of this network was considered to be made in the shape of a cube, which therefore turns out to be the shape most perfect and balanced shape that can be obtained from static geometry, while the sphere is the most perfect and balanced shape that can be obtained from dynamic geometry, see: **Hecht, 2004**, which to represent it requires the development of an Algebra of Nature, (see Chapter 9), suitable for representing the evolutionary dynamics inherent in the intimate nature of the Universe, composed of an "*Essence*" that the ancients called "*Ether*", or Quantum Void having the properties of neural superfluid, see: **Scala 2022a**.

5) Sense of Limits in Universal Forms

For classical science, all systems existing in nature are considered as separate machines, made up of physical and chemical processes, but this is not enough to explain the various "*whys*" of their existence and their forms; where instead for the "Science of Resonances", the "*state of Universal imperfection*", see: **Ivaldi, Smigliani, 2024a**, generates the "*sense of the limit*", at the basis of the Symmetry and Topology present in all the "*Universal Harmonic Forms*"; from which, for example, living beings, thanks to their forms, acquire awareness of themselves, and therefore of existing.

The "*sense of the limit*" in the micro-cosmos is found in the Planck length, see: **Bolotin, Tour, Yanovsky, 2020**; while in the macro-cosmos, we find the "sense of limit" in the size of the Universe, which is enormous (therefore measurable), but not infinite, see: **Ord, 2021**; and we also find it present in the speed of information, which is enormous but not infinite, and should correspond to the speed of Light raised to the fortieth power (yet to be demonstrated in the laboratory). In mathematics, the "*sense of the limit*", capable of demonstrating that every non-trivial representation system (i.e. capable of representing the arithmetic of integers), is incomplete and/or incorrect, we find it both in Gödel's incompleteness theorem, see: **Beklemishev, 2011**, and both in the Riemann hypothesis, see: **Smigliani, Ivaldi, 2024**.

The "*Torus geometric shapes*" present throughout the Universe, see **Fig. 2**, from DNA, to the elegance of galaxies, see: **Pletser, 2017**, have harmonious perfection in common, not only as regards the concept of awareness, but also for their functional motivations; for example, the stomach has a sac shape, since this is the ideal structure to carry out its task.

We note here, that already in several ancient cultures, the role of "*forms*" in their life was present; see for example the sacred geometry of the Platonic solids, which was thought to represent the

basic elements on which the entire Universe was built, and which we find in modern times in the hypothesis on the geometry of the atomic nucleus, by Professor Robert J Moon, see: **Hecht, 2001**.

Among the forms of sacred geometry, one of the best known is the one called "*Vesica Piscis*" (mystical almond), see: **Sparavigna, Baldi, 2016**; deriving from the intersection of two circles having the same radius, with the circumference of one passing through the center of the other and vice versa. The two segments that are created within this figure, one that crosses the center horizontally, and the other that vertically connects the points of intersection of the two circles, stand between them in a relationship congruent with the golden proportion ; which derives from a rectangle divisible by a segment into a square, and into a smaller rectangle, having the same ratio between the sides as the first.

This proportion, called the golden number: 1.618; represents the ideal relationship between two lengths, capable of producing a harmonious aesthetic effect, present in many natural organisms, such as the spiral shape of the Nautilus Pompilius, and everywhere in the Universe, see: **Fig. 2**.

The characteristics of this number are also present in the Fibonacci sequence, see: **De Santis, Ivaldi, 2021**, in the proportions of many sacred structures, explored in depth in 1509 by the mathematician Luca Pacioli, see: **Ciocci, 2017**, and used by Leonardo da Vinci in his artistic creations.

The figure of the "*Vesica Piscis*", we also find it present in the figure of the "Flower of Life", present in ancient finds that emerged in various areas of the Earth, such as: Egypt, Ireland, Turkey, England, Israel and China. According to researcher **Manninen, 2016**, the structure of the "*Vesica Piscis*" recalls the division process, thanks to which the cell, and consequently life, originates.

Interesting are the functions inherent in the pyramidal shape with a square base, which represents the Octahedron in the Platonic solids, and at the same time can have the function of a transmitting antenna, both for the fundamental frequency and for multiple frequencies, see: **Minin, Minin , Yue, 2020**. Furthermore, the Pyramid emits scalar waves in a North-South direction, by means of a halo of colored rays not visible to the human eye.

Perfect geometric and symbolic structures are also present in the "mandalas" used in the Tibetan, Nepalese and Indian traditions; constituting tools of initiation and access to higher states of consciousness, which can be activated through contemplation.

Finally, for those who study the life sciences, see: **Delplace, Marston, Venaille, 2017**, the questions regarding heredity and the properties of living organisms are interesting; in classical science they are attributed to probable functions of DNA that are still unknown; where biologists call "morphogenesis", see: **Sheldrake, 2006**, the sector of studies concerning the origin of form (from the Greek: morphé, form, and genesis, origin), try to give explanations by relying on genetic programming, where each species does nothing but follow the instructions of its genes, and where genetics does not yet provide clear explanations regarding the mechanism underlying the transition from "*egg*" to its adult form in living organisms.

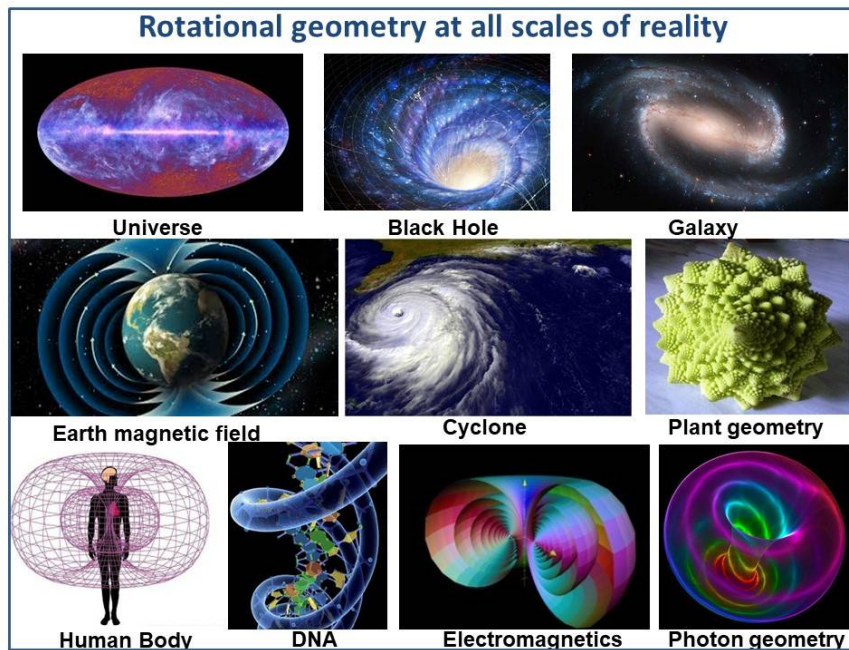


Figure 2: Geometry of the “Torus”, present on various scales in the Universe

6) Micro-structure of Space, and Polarity of Charge

For the "Science of Resonances", Space has a micro-structure in the shape of spherical micro-bubbles of potential energy, where no bubble moves very far from its position, as happens to a group of objects floating on the surface of the sea, so every time a stimulus reaches a field bubble, the impulse will be transferred around, even if the bubbles continue to maintain the same spatial position. In this primordial Ocean of field bubbles, when through a frequency such as a photon, see De Pretto in: **Ivaldi, Meijer, 2022**, a stimulus is generated in an area of Space, in particular resonance conditions atoms can be generated in the shape of vortices similar to smoke rings or eddies, which through a form of Universal Intelligence, see: **Meijer and Ivaldi, 2022**, which David Bohm identified as "*Implicated order*", forms within the Quantum Void, or Space with neural superfluid characteristics, the harmonic Universal geometric forms, see: **Fig. 2**.

From this fact it is clear that the concreteness of matter is only our impression deriving from the game of mirrors of the apparent "*Universal Duality*", furthermore matter can always be converted into energy and vice versa, through the famous equation $E = mc^2$, see: **Schuchewytsch, 2019**; so the observed teardrop shapes of the electron are exactly what we would expect to see by studying a vibratory wave at rest, which evolves taking on the geometric shapes of the "*Platonic Solids*", already investigated by the US physicist R. J. Moon, see: **Hecht, 2001**, which have been of great importance since the times of the ancient Sacred Science, and which appear naturally in a spherical vortex of vibrating and pulsating fluid, see: **Lindberg, 2012**.

We can conclude here that atoms are vortex formations, in which the "*Polarity of the charge*" is seen as the etheric pressure that is directed towards the center of the spherical field, which can also be interpreted as the struggle of matter and energy to return to the One embryonic Perfect primordial.

7) There is no separation between Space, Matter and Consciousness

The "Science of Resonances" directs us to accept the fact that the so-called "*Condensed Matter*" is nothing other than "*Condensed Space*", and consequently the atoms and molecules are seen as the flame of a candle, in which the energy it releases (such as the heat and light of the flame), must be balanced by the energy it absorbs (such as the candle wax and the oxygen in the air).

This "*Universal Breath*" analogous to the candle is a distinctive feature of Dr. **Puthoff's 2003** model of the atom, with which he tries to explain why the hypothetical electron does not radiate all its energy around and falls into the nucleus.

This perpetually pulsating "*Universal Breath*", see: **Lindberg, 2012**, finds analogies with the work of the great Russian scientist Nicolai Kozyrev, see: **Bischof, Ludwig, 2020**, who tends to visualize all physical objects as sponges immersed in water, and this is just one aspect of his exceptional work, which shows that all fields such as gravitational and electromagnetic forces are simply effects resulting from different forms of ether/ ZPE moving in a toroidal environment.

The "Science of Resonances" also leads us to understand that we can get to the real origin of the interaction phenomena between electromagnetism and gravity only by removing the Space-Time of Relativity, and reconsidering the Superfluid Vacuum as the basic Essence of the Quantum Vacuum: Gravity would be a simple phenomenon of difference in its "*density*" (spatial ethereal density), see: **Fedi 2017**, thus arriving at the correlation between electromagnetism and Gravity, present in the "Symetrization of Maxwell's Equations", see: **Scala, 2022b**.

We note here that Einstein supported the rejection of the ether in 1910, but in 1920 Einstein stated that in reality the hypothesis of the existence of the ether does not contradict the theory of special relativity, in fact then in 1924, he wrote: "*In theoretical physics, we go nowhere without the ether, i.e. a continuum of defined physical properties, as the general theory of relativity excludes long-range direct action; and every short-range theory assumes the presence of continuous fields, and, consequently, the existence of the ether.*" From here it can be deduced that Einstein was the scientist who made the most mistakes in the world, and several times had the courage to admit his mistakes, see: **Tombe, 2020**.

We consider the concepts of Intelligence and Consciousness at a human level in a separate way, where we see Intelligence as the ability to reason, solve problems, define and study ourselves and the world in which we live; while we see Conscience as an intimate feeling, designed to make us aware of our existence. In the "Science of Resonances" these two concepts are seen on a Universal level, see: **Ivaldi, Smigliani, Scala, 2024**; where the concept of Consciousness represents the fabric of reality in an evolutionary resonant transition see: **Meijer, Marrone, Axelrod, 2023**, deriving from how a living being perceives and manages a state of information, which as an evolutionary pinnacle can lead to a state of "*Conscious Consciousness*", which Plato described as something that resides in the world of ideas, Aristotle described it as something that animates life, Galileo described it with mathematical concepts, Descartes saw it as something separate from matter, to arrive at the "Science of Resonances", which sees it as a continuously evolving flow of information connected in real time, see: **Meijer, Ivaldi, Faixat, Klein, 2021**; and where the role of Symmetry and Topology is fundamental, which are expressed through different waveforms, starting from

electro-magneto-gravitational waves, see: **Scala, 2022b**, to scalar waves, see: **Meyl, 2001**, to soliton waves, see: **Geesink, Meijer, 2016**, and **Veenstra and others, 2023**, to arrive at the semi-harmonic frequency model that organizes local and non-local states via quantum entanglement in both EPR studies and life systems, see: **Geesink, Meijer, 2018**. Note that each of these waveforms has its own precise role in the harmonic functioning of the Universe.

In this Universe where there is no separation between Space, or Quantum Void, which the Ancients saw as Ether, and therefore where Everything is connected in real time, the concept of "limit", deriving from a state of *"Universal imperfection"*, see: **Ivaldi, Smigliani, 2024a**, as a concept of measurement, represented a cornerstone of Greek thought, in which the concept of limit, *"Peras"*, for Aristotle meant: *"Extreme term of each thing, beyond which one can no longer find nothing of the thing, and beyond which there is the whole thing. Limit is the greatness of a form. Limit is the purpose of existence for each thing. Limit is the essence of each thing, understood as the limit of knowledge, and if it is the limit of knowledge, it is also the limit of the thing"*.

8) Role of Apparent Universal Duality, Degree of Judgment in Conscious Life, and Existential and Development Model Crises

The game of the apparent Universal Duality, see: **Faixat, 2011**, has the function of a *"mirror"*, which generates the illusion of the *"concreteness of condensed matter"*, and of the flow of Time, deriving from the oscillation or pulsation present everywhere from micro to the macro cosmos, see: **Lindberg, 2012**, and also: **Pletser, 2017**, which ranges between the universal primordial Perfect One, which above resides in the sixth dimension, and below towards the **0.5**. Exactly in the middle we find the *egg*, where the generating Power of being is concentrated, which divides into + and - to be born. With this separation, the generating Power is fragmented into a harmonious and balanced polarity, where the power of the individual is not in conflict with the power of the other, but both feed and support each other, in harmonious resonance with the *"Evolutionary Arrow"* proper to the Universal Consciousness. At the level of Intelligent and Conscious intellectual process (inseparable for the Science of Resonances), see: **Ivaldi, Smigliani, Scala, 2024**, the power of being, we find it in the orbital of the electron, which can be compared to information, identifiable by means of two *"judgements"*, having the same energetic weight, but with opposite sign (power +/-), where in addition to having a +/- sign, they have a different energetic weight, which increases with increasing distance from the *"Evolutionary Arrow"* of Universal Consciousness.

To understand the function of the *"judgment"* element, it is necessary to immerse ourselves in a vision where in this Universe, which *"appears bipolar"* to us, the energetic weight of the *"judgment"* in relation to the resonance with the *"Evolutionary Arrow"* of Universal Consciousness, can be interpreted as: everything that *"unites"* goes towards the + sign, (where the saying applies: *"harmonious beauty will save the world"*), while instead everything that *"divides"* goes towards the - sign, (where the said: *"divide and conquer"*).

It should be noted that the degree of judgment can only be triggered when the state of *"conscious consciousness"* appears in the various processes of life, which is based in the fifth dimension (or

pulsating flow of information, from which free will arises). For example, in the state of life inherent in the era of dinosaurs, the degree of judgment could not be present.

The "*power of judgment*" element finds its reason for existing only if we admit the existence of a "*generating entity*" in the sixth dimension, which can manifest itself in the natural flow of certain areas of our lives. As a metaphor we could imagine that finding ourselves floating inside a large tank, where, through a tap, the water flows gradually allowing us to rise uniformly, if at a certain point we notice that this flow changes in intensity, this fact indicates to us that outside there is it is an "*entity*" that operates on the tap. This "*entity*", or Primordial Perfect One, based in the sixth dimension, is the one that establishes the "*degree of judgment*", deriving from the free will present in conscious life, where:

A) When it resonates towards the - sign (where our behavior goes towards what divides), it triggers a downward spiral that can lead to the cancellation of the state of being.

B) When it is in resonance towards the + sign (where our behavior goes towards what unites in Universal Harmony), it triggers the evolution of life towards the Perfect One, where every form of life contributes to embellish and enrich its garden, in particular those that have developed a state of Aware Consciousness, which constitute the most noble and beautiful part of it.

The "*degree of judgment*" should not be confused with what we identify with a thousand facets as *good* and *evil*; since the Universe has no feelings, it is "**Harmonically right**" based on simple and inexorable rules, such as the Law of simplification, also called "Occam's Razor Law", and the Law of Resonance inherent in the Intimate Nature of 'Universe, which as described in **Fig. 3** always and only points upwards, towards the Perfect One, which also being a "*Generating Power*", can intervene in the various evolutionary processes. From these simple Laws, from which nothing and no one can escape, it is clear that in our journey through imperfection see: **Ivaldi, Smigliani, 2024a**, through our precious errors, we are destined for evolution.

The "*Evolutionary Arrow*" that pushes everything upwards, see: **De Santis, Ivaldi, 2021**, can be mathematically deduced from the Fibonacci-Lucas sequence, **Fig. 3**, where an infinite series of numerical sequences point to "*Perfection*"; since as the value of m and n (column and row number) increases, the ratio of a term $S_{m, n}$ with its subsequent $S_{m, n + 1}$, gets closer and closer to the Perfection Ratio ϕ . We are aware here that: when with increasing position terms, the difference between the Ratio $S_{m, n} / S_{m, n + 1}$ is ϕ , turns out to be less than the minimum dimension referring to our reality (Planck length), we are in the sixth dimension where resides the Primordial Perfect One, who if he generated a perfect creation, could only generate a clone of himself, and that would be meaningless. Therefore the Primordial Perfect One, to generate new experiences, had no other choice than to pass through the ballot present in a state of imperfection, which: "**represents the price we must pay in order to exist**", see: Pg.5 of the article: **De Santis, Ivaldi, 2021**, from which it is clear that enjoying our freedom lies precisely in that imperfection below the limit of the Perfect, where the possibility of self-repair also exists.

n	m	0	1	2	3	4	5	6	7	8	9	10	11	12	13
30		832040													
29		514229	1149851												
28		317811	710647	1589055											
27		196418	439204	982090	2196020										
26		121393	271443	606965	1357215	3034825									
25		75025	167761	375125	838805	1875625	4194025								
24		46368	103682	231840	518410	1159200	2592050	5796000							
23		28657	64079	143285	320395	716425	1601975	3582125	8009875						
22		17711	39603	88555	198015	442775	990075	2213875	4950375	11069375					
21		10946	24476	54730	122380	273650	611900	1368250	3059500	6841250	15297500				
20		6765	15127	33825	75635	169125	378175	845625	1890875	4228125	9454375	21140625			
19		4181	9349	20905	46745	104525	233725	522625	1168625	2613125	5843125	13065625	29215625		
18		2584	5778	12920	28890	64600	144450	323000	722250	1615000	3611250	8075000	18056250	40375000	
17		1597	3571	7985	17855	39925	89275	199625	446375	998125	2231875	4990625	11159375	24953125	55796875
16		987	2207	4935	11035	24675	55175	123375	275875	616875	1379375	3084375	6896875	15421875	34484375
15		610	1364	3050	6820	15250	34100	76250	170500	381250	852500	1906250	4262500	9531250	21312500
14		377	843	1885	4215	9425	21075	47125	105375	235625	526875	1178125	2634375	5890625	13171875
13		233	521	1185	2605	5825	13025	29125	65125	145625	325625	728125	1628125	3640625	8140625
12		144	322	720	1610	3600	8050	18000	40250	90000	201250	450000	1006250	2250000	5031250
11		89	199	445	995	2225	4975	11125	24875	55625	124375	278125	621875	1390625	3109375
10		55	123	275	615	1375	3075	6875	15375	34375	76875	171875	384375	859375	1921875
9		34	76	170	380	850	1900	4250	9500	21250	47500	106250	237500	531250	1187500
8		21	47	105	235	525	1175	2625	5875	13125	29375	65625	146875	328125	734375
7		13	29	65	145	325	725	1625	3625	8125	18125	40625	90625	203125	453125
6		8	18	40	90	200	450	1000	2250	5000	11250	25000	56250	125000	281250
5		5	11	25	55	125	275	625	1375	3125	6875	15625	34375	78125	171875
4		3	7	15	35	75	175	375	875	1875	4375	9375	21875	46875	109375
3		2	4	10	20	50	100	250	500	1250	2500	6250	12500	31250	62500
2		1	3	5	15	25	75	125	375	625	1875	3125	9375	15625	46875
1		1	1	5	5	25	25	125	125	625	625	3125	3125	15625	15625
0		0	2	0	10	0	50	0	250	0	1250	0	6250	0	31250

Fibonacci	$F_n = S_{n-0,n} = \frac{\Phi^n - (-1)^n \times \Psi^n}{\Phi - \Psi};$
Lucas	$L_n = S_{n-1,n} = \Phi^n + (-1)^n \times \Psi^n;$

Figure 3. The sequence of the first 11 Fibonacci - Lucas sequences obtained from the numerical sum of the first 31 terms of the Fibonacci sequence.

Here we become aware that the first five dimensions are of an "operational nature", which on the basis of statistical science, could justify a "static Universe"; while to justify the "Evolutionary Arrow", at the basis of the birth and evolution of Universal Life, requires the presence of a sixth dimension from which the generative and management inputs of the "Everything" start, where living beings who have reached the state of Consciousness Conscious, with their apparent "free will" they can only evolve in upward Resonance, otherwise if they try to travel the Resonances that point downward, they inevitably go towards their self-destruction.

For us living beings who have reached a state of Aware Consciousness, alongside the primordial instincts for nutrition and reproduction, in which the principle of selection of the most adaptable is in force (which modern society transforms "into the strongest", in contravention of the true words of Darwin, in which the first is praised and the last relegated to oblivion), other values should overlap, such as the appreciation and valorisation of the Universal harmonious beauties of which we too are part, accepting responsibly and in "resonance" the simple and inexorable Universal laws and respecting Nature with less aggressive behaviors.

Instead we observe that unfortunately a lot of humanity is still more linked to the life philosophy of **Having** than to that of **Being**, directed towards the development of our highest resonances, and this behavior of ours is also at the basis of the existential crises especially present in the most young people (since the profound reason why they appeared in this Universe and what their final destination is has not been explained to them); and it is also at the basis of the current development model crises, such as that of "Climate Change", largely deriving from the great energy need to fuel our current technological development model, which in any case was only possible for a part of humanity; and therefore it appears increasingly clear that if we continue to

use this development model, there will not be enough resources on Earth to allow industrialized countries to continue to maintain a high standard of living, and at the same time allow poorer countries to develop decently.

The issue can only be resolved by innovating the foundations of our current Science, in which for example the "Science of Resonances" will lead us towards a paradigm shift in the very concept of energy, see: **Ivaldi, Smigliani, 2024c**, and **Hotta 2010**.

The various crises that are gripping humanity on this Earth, if we analyze in depth, we discover that they are induced by a **"basic materialist science"**, which divides everything, starting from the so-called condensed matter, which for classical science is composed of *"concrete and separate particles"*, suspended in an *"empty space"*, as required by Einstein's Relativity; while the "Science of Resonances" teaches us that nothing is separable in this Universe, and this concept is fundamental to make us understand what we are, and what we are doing on this enormous and Harmonious Brilliant Garland called the Universe, where everything is life in continuous evolution, and as De Andrè sang: *"Flowers are born from manure"*, and death is pure illusion, see: **Meijer, 2022**.

9) The Role of Resonances in the Algebra of Nature, at the Base of Universal Life

Before addressing the topic inherent to the role of Resonances in the Algebra of Nature, it is good to observe that in classical Tensor Algebra, a *"Tensor"* is seen as the representation of any quantity, physical or geometric, which has an existence independent of the reference system chosen to represent them; therefore the components of a tensor in a given reference system are linked to the coordinates of the tensor itself. This property is the basis of the rules of classical tensor calculus, in which a scalar entity is understood as a physical quantity defined only by a number at each point in space, such as the temperature, which at each point in space is given by its size, and where a vector is defined with a number of components equal to the dimension of space, so on a plane a vector has two components, in space it has three, in space-time it has four. In classical mathematics, therefore, a tensor is the generalization of the concept of scalar and vector, from which it can be deduced that a tensor field can have multiple components and act on multiple spaces.

We note here that the first use of the tensor concept occurs in the context of continuum mechanics, in connection with the need to describe the stresses and deformations undergone by extended bodies, hence the formalization of rational mechanics.

To describe the role of Resonances in the evolutionary mechanisms present in the intimate Nature of the Universe, an *"Algebra of Nature"* is needed, where it is clear that Universal Nature is not as complex as our classical science describes it, which divides everything into a thousand specialized streams that make him lose the sense of the *"vital principle"*, in which the "Science of Resonances" teaches us that everything began from a low generating frequency of **50 Hz ($\pm 10\%$)**, see Chapter **11**), intended not as an error but as a limit of information from which free will arises once the awareness of existing is acquired thanks to the apparent duality: $\pm$.

In this context, for example, **England, 2022**, considers the phenomenon of Universal life as an *"organized and resonant"* system, with a basic *"driving"* energy, in an environment capable of retaining heat. A new concept for thermodynamics, where entropy represents a return of energy to the environment, and not an irreversible dissipation. In other words, a living system would become such thanks to the ability to *"use"* the energy available and not to *"suffer"* it as an inert and therefore closed system would do, as foreseen in the second law of thermodynamics: in practice a concept close in **Prigogine, 1984**.

The behavior of the proteins that support information in our organism shows us that reality is interconnected along an exponential spiral matrix, see: **De Santis, Ivaldi, 2021**. Proteins, including the side chains of amino acids, have a helical component predominantly right-handed, but also left-handed, with the ability to perform their binding function in real time, like a spatial tensor, or like a tunnel effect, see: **Smigliani, 2022**.

The possibility of combination in the three-dimensional context that proteins have cannot be left to chance, since they are formed in resonance on an electromagnetic grid that oscillates in a space of frequencies, between 0.1 Hz and 2 MHz, see: **Damascus, Giuliani, 2017**. We note here that: in nature there are resonant frequencies with some of our proteins (see: **Fig. 4**), such as the Schumann frequency of 7.8Hz, which resonates with the hippocampus of our brain. These helical proteins are part of the reticular structures of the *"cellular matrix"* and are similar to those that make up the cytoskeleton of cells, as well as the tubulins that interconnect neurons.

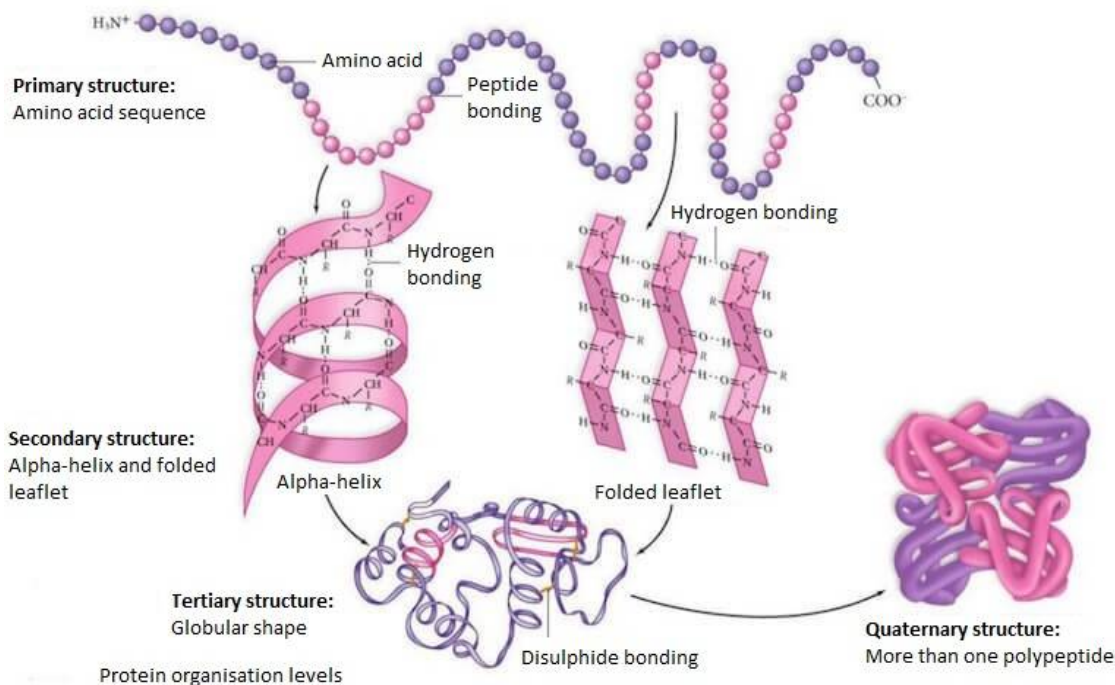


Figure 4 - Image downloaded from *thechimica-online.it* website, the secondary and tertiary structure of proteins.

Taking advantage of the concepts set out above, let's imagine a journey into the fourth spatial dimension which is beyond our senses and which does not depend on Time being non-local in nature. To avoid confusion with the four dimensions contemplated by Minkovski's relativistic space-time, we could call this fourth spatial dimension the **"5th dimension"**.

Now let's imagine describing the motion of a bit of information in this dimension by exploiting the hourly equation of motion, doing without time:

$$\mathbf{S}(t) = \mathbf{V} \cdot (t - t_1) + \mathbf{S}_i \text{ that with } t = 0 \rightarrow \mathbf{S}(t) = \mathbf{S}_i$$

Therefore information would be omnipresent on the entire surface of the 5th dimension. If this were the case, it would be absolute perfection where everything is still (like Parmenides' man) and where everything is immutable. This consideration should therefore not contemplate the birth and development of *"Life"*, whereas we know that this is not the case. It is obvious, then, that at a certain point this state of perfection is interrupted by a *"pulsation"* capable of creating an asymmetry or an *"error"* in its fluid immobility; a pulsation that brings with it information ready to *"create something different"*. At this point we can therefore say that **t** could be replaced by **p** (understood as impulse, pulsation), so:

$$\mathbf{S}(t) = \mathbf{V} \cdot (t - t_1) + \mathbf{S}_i \rightarrow \mathbf{S}(t) = \mathbf{V} \cdot (\mathbf{p} - \mathbf{p}_1) + \mathbf{S}_i$$

Since we said that the 5th dimension has a non-local nature, the velocity **V**, lacking time, cannot be represented by the relationship between space and time, but is replaced by two non-local quantum states (i.e. entangled), **S_i** and **S_i'**:

$\mathbf{V} = [|\mathbf{S}_i\rangle + |\mathbf{S}_i'\rangle]$, so the hourly equation becomes

$$\mathbf{S} = [|\mathbf{S}_i\rangle + |\mathbf{S}_i'\rangle] \cdot (\mathbf{p} - \mathbf{p}_1)$$

The term **S_i** disappears as expressed by the two quantum states, and **S** indicates the Space or Quantum Vacuum and where

$$(\mathbf{p} - \mathbf{p}_1) = \text{pulse interval through which information propagates}$$

Following Occam's razor rule (simpler and energetically favorable solution).

So long as $\mathbf{p} = \mathbf{p}_1 = \pm 0,5$, (zero point energy¹: $\mathbf{E} = \frac{1}{2} \hbar \mathbf{v}$), we have that:

$$\mathbf{S} = \{[|\mathbf{S}_i\rangle + |\mathbf{S}_i'\rangle] \cdot (\pm 0,5)\}$$

Which represents a pulsating wave whose complete oscillation follows a harmonic trend, so (**Fig.5**)

1. In the case, for example, of the harmonic oscillator, which constitutes a useful paradigm for a large number of physical systems, the energy corresponding to a generic state containing n quanta of energy at frequency ν is $E_n = h\nu(n+1/2)$, where h is Planck's constant. From this it can be deduced that, even in the absence of quanta ($n=0$), the system remains the so-called zero point energy, equal to $E_0 = h\nu/2$. From "Treccani enciclopedia online: https://www.treccani.it/enciclopedia/vuoto-quantistico_%28Enciclopedia-Italiana%29/

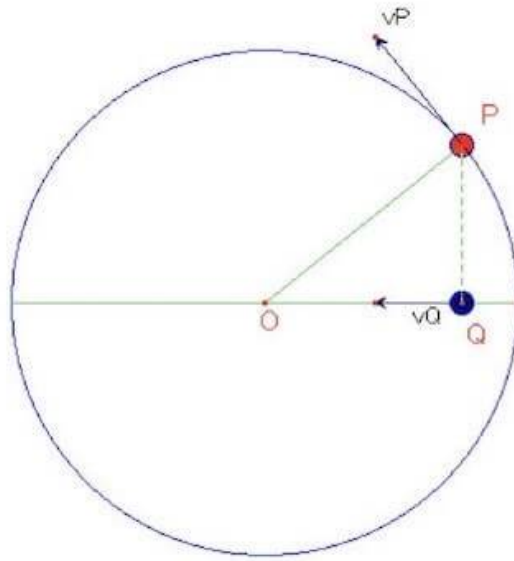


Figure 5. Harmonic movement (<https://www.chimica-online.it>)

Whose hourly equation:

$$X(t) = r \cdot \cos(\omega t)$$

following the reasoning made previously, it becomes:

$$S(p) = r \cdot \cos(\omega p)$$

From which some considerations can be deduced, also by observing the graphic design:

- 1) The oscillation period **T** (complete revolution), being quantum in nature, recognizes predetermined quantum jumps at: $\pi/2 (+ 1/2)$, $\pi (+ 1)$, $3\pi/2 (- 1/2)$ e $2\pi (\pm 0,5)$;
- 2) The completion of the revolution, however, does not make the initial state coincide with the arrival point **2π** , since the impulse would cancel out: every quarter of a revolution the impulse in longitudinal spiral advancement curves by a value equal to the golden number, according to the progression of Fibonacci creating an advancement of the impulse.
- 3) The meeting of two of these waves is capable, through the sum of their potential energies, of releasing kinetic creative energy (**mc^2 or mv^2** , as anticipated by Olinto De Pretto a few years before Einstein) subject to the **"Time in which it expresses itself"** (beginning → arrow of time → end and return to the initial potential energy); this energy will be pulsating (and therefore capable of acquiring a consciousness: organic life) if in phase coherence, helical and with opposite directions, more stable and structured (therefore not capable of acquiring a consciousness: lithosphere) if in phase coherence helical with the same direction (see Palmieri: **Ivaldi and Meijer, 2022**)

As can be imagined, a longitudinal wave propagation with the characteristics of the Fibonacci spiral forms a toroidal rather than classical Riemannian geometry.

The cosine of the impulse at its origin (see: **Fig. 5**) will also have a path in length dependent on the radius itself of the oscillating period **T**: in other words if the radius is larger, the further path the cosine will have to travel to allow "*complete oscillation of the impulse*". Given the quantization of the progression of the impulse (like a loose thread of a rosary), the measurements of the radius, therefore the amplitude of the oscillation, are also quantized.

If we consider the pulse as a Photon, the meeting of two pulses (two photons) will lead to the fusion of their two potential energies, releasing kinetic energy

$$\frac{1}{2} m_1 c^2 + \frac{1}{2} m_2 c^2 \rightarrow E = mc^2, \text{ with } m_1 \text{ e } m_2 \text{ understood as } \textit{virtual masses}$$

The resulting "*mass*", therefore, which forms an electron, is simply a "*condensation of potential energy into kinetic energy*", from which it is clear how the information contained in the 5th dimension can give rise to reality as we do it. we know.

The singularity is that this "*condensed mass*" to then go on to build the reality of particles, at this point understood as "*condensed energy packets*", will necessarily have to form "***geometric shapes capable of resonating***" with each other in order to build "*matter*", both inert and living. Applying the Fibonacci progression to geometry, these solids are the Platonic ones: tetrahedra, hexahedrons, octahedra, dodecahedrons, etc.

Given that the first atom to condense in Space from the 5th dimension is hydrogen, it is plausible that it is the one more capable than the others of oscillating in perfect phase resonance with the impulse/information of the 5th dimension itself.

Being made up of only a proton and an electron (without the neutron which would weigh down its movement), the latter is able to jump on 7 quantized levels of energy and six sub-levels called "***qubits***": in total 13 quantum states capable to oscillate in resonance with each other and with other atoms.

The resonance oscillation produced by the mass condensation of energy, therefore, creates "time", understood as an evolutionary arrow advancing from a point **t₁** (appearance of the mass) to time **t₂**

(return of mass into energy). Time therefore creates speed which, however, syllogistically, cannot be constant: if the basic speed associated with **r = 1** (energy state of the first level of the hydrogen atom, oscillating at the minimum energy equal to **0.5**) is that of light, for other values of **r** this speed increases, since it would depend on the energy density of Space (to be demonstrated in the laboratory).

If we consider, in fact, that the zero point energy, in the fifth dimension, does not refer to time as a measurement (**t = p = ± 0.5**), but travels in entanglement as information, then the need to measure the speed disappears in the space; in this way the relationship **E** (energy) and **m** (mass) varies in relation to the "*density of the energy field in Space*", which we could call **τ_s (Tau_s)**:

$$E/m = \tau_s.$$

Considering $\tau_s = 1$ (reference value that indicates the dimension in which we exist, the Earth), we will have:

$$E/m = 1$$

That is, perfect balance between the energy trapped in the mass and the free potential energy in the quantum vacuum (Space), accounting for the fact that the formula of special relativity describes only a part of Space, that is, the temporality of the condensation of energy into mass in our size, with $\tau_s = 1$. The apparent duality that appears in the condensed dimension ($\pm$, matter/antimatter etc) explains that $\pm 10\%$ described in relation to the base frequency of **50 Hz**: oscillation which is not "*the error in the classical sense*", but something that it allows life to choose to "*evolve*", in continuous connection with Universal Intelligence (remember that for the Science of Resonances there is no difference between Intelligence and Consciousness).

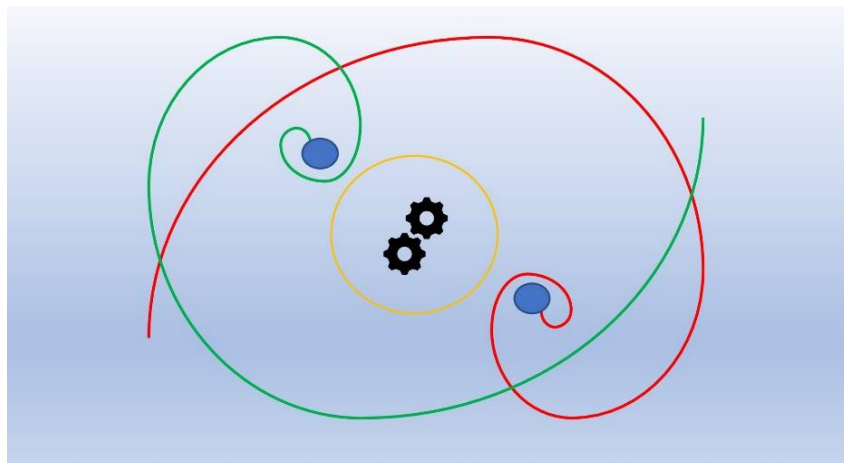


Fig. 6. Digital drawing on hard copy, property of Nicola Smigliani

Background: represents the sixth dimension, the Cosmic One, permeating a fluid and perfect Universe, a 'superfluid'.

[●] : example of an impulse generated entangled in two points separated by the Universe, with opposite spiral progression.

Green spiral: **left-handed** Fibonacci spiral that behaves like a **solitonic wave** with immutable characteristics entangled with Space (quantum vacuum) with base energy ($E_{base}=0.5$, **ZPE**). Time is not of the essence.

Red spiral: **right-handed** Fibonacci spiral that behaves like a **solitonic wave** with immutable characteristics entangled with Space (quantum vacuum) with base energy ($E_{base}=0.5$, **ZPE**). Time is not of the essence.

[⚙] : the meeting of two **solitonic waves** (or pilot waves, or «**photons**») creates a **turbulence** which has a «beginning» and an «end» ($T1 - T2$, **relativistic space-time place**), within which the atomic particle electron (negative **green** and positive **red: matter** and **anti-matter**), which are actually two sides of the same coin: the existence of a single 'dual' matter, (+-), from which derive (by further meeting of waves generated by the first two generators) the proton, the neutron and the first condensed atom, Hydrogen; to follow all the other elements of Mendeleev's table.

From this point of view it is clear that algebra as we conceive it is not able to describe Nature, indeed, at times, it tends to make it more complicated than it is.

The Algebra of Nature, in fact, frees itself from the concept of zero understood as nothing, recognizing the fact that everything that evolves in the Universe starts from **0.5** (see: **Ivaldi, Smigliani, 2024a**), and evolves with a similar progression to that of Fibonacci, creating geometric solids capable of resonating with each other, as does the algebra of hydrogen bonding, see: **Smigliani, 2023**, which describes a "*pulsation*" that translates into bits of information resonant on 13 Quantum Spaces : the seven levels allowed to the electron and the six spaces (qu-bits) between one level and another.

Thanks to hydrogen bonding, this "*pulsation*" mode, see: **Lindberg, 2012**, is transmitted to every overlying molecular group (provided it combines in resonance with the Carbon²), until at a certain point, a sufficiently complex organism acquires the "*consciousness*" of this pulsation (*life*), structuring its frequencies in a coherent and resonant way, and thus acquiring the ability to write one's temporal experience on the fabric of the neural Superfluid Quantum Void, see: **Scala, 2022a**. Here a further singularity arises: the 13th note, corresponding to the seventh energy level, shows borderline behavior, as if on the one hand it was attracted by the Quantum Void or Space, while on the other it was trying with "*all its strength*" to release energy to return to the energy balance level of hydrogen: a seemingly bipolar characteristic.

At this point the two further frequencies would appear which would close the palindromic progression of Moody 8 and 9, see: **Ivaldi, Moody, 2021**, giving life to two new notes (to be confirmed in the laboratory), an octave higher than the first energy level (harmonic of **C** in the natural scale, centered on the absolute A) and an octave higher than the first space (harmonic of the first qubit).

By further posing the condition that the first energy level of hydrogen resonates with the base frequency of space (*0.5 Hz with pulsation 3.14*) and does so (see Chapter 11) by generating a frequency equal to $0.5 \times 10^2 \text{ Hz}$, with pulsation equal to 3.14×10^2 , points of contact with the natural scale of sound (according to Western tradition) emerge:

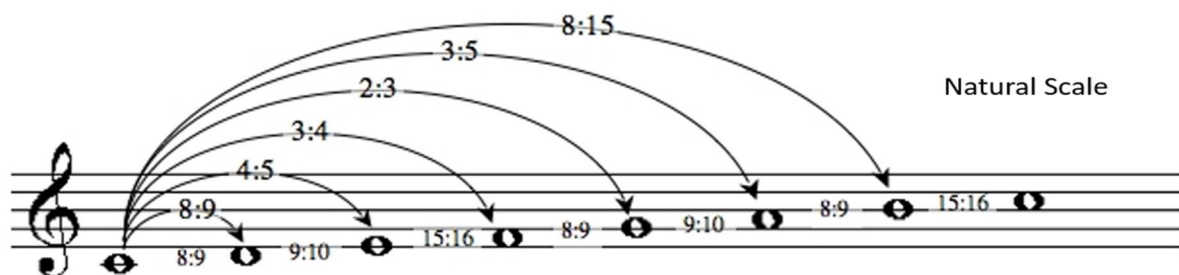


Fig. 7. Image taken from Wikipedia

2. Carbon and Silicon are the two atoms with which hydrogen "resonates" in perfect phase, similarly to what it does with Oxygen. Therefore it traces two paths: that of Carbon (plastic and ductile like the hydrogen bond itself) which goes towards the living organism, that of Silicon (rigid, structured and resonant with itself), towards the so-called inert matter

Where the frequency of the **C** is a harmonic of **50 Hz** ($\pm 10\%$): 264 Hz (250 Hz $\pm 10\%$), as well as the absolute **A** (450 Hz $\pm 10\%$), both in the “*Standard*” version with the **A** at (440 Hz), which in the “*Golden Section*” one built on the Fibonacci sequence with the **A** at (432 Hz). The nature of the two scales centered on the different frequencies of the absolute **A** was addressed in **Ivaldi, Smigliani, 2024a**, in relation to the algebra of the hydrogen atom based on the energetic quantum jumps of its electron, where the possibility of further notes was highlighted to be added to those already existing in the classical pentagram; to be precise, a further 6 in the “*Standard*” scale with **A** at 440 Hz (for a total of 13), and another 2 in the “*Golden Section*” scale, with the **A** at 432 Hz:

“Standard” scale (La = 440 Hz)

DO = 264 Hz

Space (qubit) between DO and RE = $264 + \frac{1}{2} (297-264) = \mathbf{280,5}$ (*DO# 272,5*)

RE = 297 Hz

Space (qubit) between RE and MI = $297 + \frac{1}{2} (330-297) = \mathbf{313,5}$ (*MI_b 313,9*)

MI = 330 Hz

Space (qubit) between MI and FA = $330 + \frac{1}{2} (352-330) = \mathbf{341}$

FA = 352 Hz

Space (qubit) between FA and SOL = $352 + \frac{1}{2} (396-352) = \mathbf{374}$ (*FA# 363,4*)

SOL = 396 Hz

Space (qubit) between SOL and LA = $396 + \frac{1}{2} (440-396) = \mathbf{418}$ (*Sol# 408,8*)

LA = 440 Hz

Space (qubit) between LA and SI = $440 + \frac{1}{2} (495-440) = \mathbf{467,5}$ (*SI_b 470,9*)

SI = 495 Hz

Space (qubit) between SI and DO in the upper octave = harmonic of the first qubit = **561**

DO in the upper octave = **528 Hz** (harmonic of the first level).

** Taken from Wikipedia.org*

“Golden Section” scale (La = 432 Hz)

DO = 256 Hz

Space (qubit) between DO and RE = $256 + \frac{1}{2} (288-256) = \mathbf{272}$

RE = 288 Hz

Space (qubit) between RE and MI = $288 + \frac{1}{2} (320-288) = 304$

MI = 320 Hz

Space (qubit) between MI and FA = $320 + \frac{1}{2} (344-320) = 332$

FA = 344 Hz

Space (qubit) between FA and SOL = $344 + \frac{1}{2} (392-344) = 368$

SOL = 392 Hz

Space (qubit) between SOL and LA = $392 + \frac{1}{2} (432-392) = 408$

LA = 432 Hz

Space (qubit) between LA and SI = $440 + \frac{1}{2} (480-432) = 456$

SI = 480 Hz

Space (qubit) between **SI** and **DO** in the upper octave = harmonic of the first qubit = **544**

DO in the upper octave = **512 Hz** (harmonic of the first level).

** Taken from Wikipedia.org*

Considerations:

A) All differences between the frequencies of the natural scale (upper frequency minus lower frequency, e.g., D minus C, E minus D, etc.) are multiples of the **prime number 11 (33, 33, 22, 44, 44, 55, -33)**; these frequencies are **different from those of the "tones"** (see in parentheses: calculation made on prime numbers, right chromatic semitone), so they would be **"new notes"**; the last difference is **"negative,"** this could trigger a pulsation (+-) such that would lead to a new natural scale.

B) The "Golden section" scale shows different frequencies from those of the natural scale, and the differences between the two notes are **multiples of the number 8 in the Fibonacci series (32, 32, 24, 48, 40, 48, 64, -32)**; these frequencies coincide with those calculated for the tones; only **two new frequencies** appear, those **between E and F and between B and C in the major octave**: therefore the **new notes** would be **only 2**? If this, were the case, then the nine frequencies recorded for the hydrogen atom, later taken up by Moody, would be justified (see: **Ivaldi, Moody, 2021**)

The only equality between the two scales is the presence of a last level (13th note in the 440 Hz scale, ninth in the 432 Hz scale) *"poised"* between remaining and going, which gives rise to a dual pulsation ($\pm$ /matter-antimatter) capable of creating a new scale of frequencies, see: **Feixat, 2011**, until completing a Mendeleev table now seen from a point of view of energetic resonances, see **Ivaldi, Smigliani 2023**:

Biostructures and Biosystems - INBB) who demonstrated how physical stimuli, such as extremely low-frequency pulsed magnetic fields, radioelectric fields, and mechanical vibration, including sound vibration, are capable of substantially modifying cell fate by orchestrating stem cell differentiation without the intervention of chemical and/or genetic agents, see: **Fontani and other, 2022**; where the cells vibrate in resonance with the Universe and receive information through electromagnetic frequencies that can be transformed into an audible sound range, tuned to and coherent with the 432 Hz of the harmonic **A** built on the Fibonacci series.

Again on the 432 frequency: it is the only tuning multiple simultaneously of both 3 and 2; genetic mathematics (sequence of active codons, interruption, other active codons and so on) follows the sequence (3;2;2;3).

Conclusion: this is not to say that music tuned to 440 Hz is less beautiful, sublime or, even, not beneficial but, simply, that tuning to 432 Hz presents an easier coherence link with the vital systems in the Universe.

10) 0.5 In Replacement of Zero = Algebraic Evidence of the “Universal Pulsation”

By equating the main equations of Classical and Statistical Quantum Physics with the new concept of zero understood as **0.5** (according to the algebra of hydrogen bonding and resonances: algebra of nature hereafter called A.N.), interesting and new questions about the true workings of the Universe arise:

In Dynamics (which studies the causes of motion of bodies), Newton's basic formula is:

$$F = m \cdot a$$

Where **F** is the resultant force (N/newton), **m** is the mass (Kg) and **a** is the acceleration (m/s²). Applying A.N. we have:

$$F - m \cdot a = \pm 0,5$$

In the area of Energy and its inescapable "conservation principle":

$$E_c = \frac{1}{2} m \cdot v^2$$

$$E_p = m \cdot g \cdot h$$

Where **E_c** is the kinetic energy (J/joules), **m** is the mass (Kg) and **v** is the velocity (m/s), **E_p** is the potential energy in a conservative gravitational field (J/joules), **m** is the mass (Kg), **g** is the acceleration of gravity (m/s²) and **h** is the height from which an object falls by gravity (m), Applying A.N. we have

$$E_c - \frac{1}{2} m \cdot v^2 = \pm 0,5$$

$$E_p - m \cdot g \cdot h = \pm 0,5$$

In Universal Gravitation

$$F_g = \frac{G \cdot M \cdot m}{d^2}$$

Where **F** is the gravitational force (N/newton), **G** is the gravitational constant (N·m²/Kg²), **M** and **m** are the masses of two bodies attracting each other (Kg), and **d** is the distance between them (m)
Applying A.N. we have:

$$F_g - \frac{G \cdot M \cdot m}{d^2} = \pm 0,5$$

In electricity

$$E = I \cdot R$$

Where **E** is volts (voltage), **I** is amperes (current) and **R** is ohms (resistance).
Applying A.N. we have:

$$E - I \cdot R = \pm 0,5$$

It is therefore evident that none of the forces described above cancels, but "*renews*" according to a "*pulsing*" logarithmic progression (± 0.5 per cycle, with $\pm \pi$ oscillation, see Chapter 11).

Particularly in Ohm's law, it is noted that the voltage (**V**, produced by a potential difference) can also "*amplify*" by an equal harmonic and resonant value and thus produce more energy than is consumed by the entire circuit to which said voltage is applied. It follows that entropy is a phenomenon closely linked to a condition of the Universe corresponding to an isolated and closed system, while outside of it can become a perfectly "*usable and recyclable*" ($\pm 10\%$?) pulsating energy, as it is the case in living beings.

The musical sequence of notes would therefore be: 12 + first qubit harmonic + C harmonic (these last two with a pulsating character, $\pm$ /matter-antimatter/stay-go). In this case the 12 fundamental frequencies of water, studied by Professor Emilio Del Giudice, see: **Ho 2015**, and those calculated by: **Geesink, Jerman, Meijer, 2020**, would find a natural connection with the fundamental frequencies of the pilot wave of the Quantum Void or Space, whatever it is.

11) Laboratory tests, to demonstrate that it is possible to communicate with space

The "Science of Resonances", see: **Ivaldi, Smigliani 2023a**, teaches us that we are part of a living and vibrant Space with which it is possible to communicate, and we have demonstrative tests carried out under a double "*Faraday cage*", to avoid any interference with the world outside the experiment. Furthermore, the experimental tests were carried out both in a normal environment and in near-vacuum conditions (to avoid interactions with atmospheric gases), in which we developed an original method (not existing in scientific literature, at least in traditional search engines) in order to induce a variation in the angle of the delta bond in order to make the graphene conductive, ***each time repeatable with the same results***.

By measuring everything with a common tester and a laboratory oscilloscope, a "singularity" emerged with the following characteristics:

- 1) Appearance of a Voltage, a Current and a Resistance which appear not to be connected to each other as Ohm's first law would require.
- 2) The voltage value is just below one volt, while the resistance, after a period of approximately 30-40 minutes, becomes almost zero, taking on the characteristics of a superconducting superfluid.
- 3) The current, with an initial peak around 6mA, after about 30-40 minutes begins to oscillate from minus to plus (+- 2-3 mA, in a slow crescendo), as if it were a "pulse".
- 4) By dividing the graphene compound into several parts, each part maintains the initial voltage characteristics: it follows that the generation of voltage energy DOES NOT DEPEND on the size and/or quantity; EVEN ONE NANOGRAM OF COMPOUND is capable of generating the same voltage. By connecting the cells in series (or the parts obtained from a single cell), the voltage adds up to half the total ($1 + \frac{1}{2} + \frac{1}{2} + \dots$ etc): in effect, a "voltage multiplier".
- 5) The choice of modified electrodes is due to the fact that they allow better detection of the effects of the experiment; and this modification does not affect the functioning of the tester, which is the same as that with normal electrodes.
- 6) On the oscilloscope, this singularity is expressed as a classic sine wave (both AC and DC); with scale X = 5 msec and Y = 2 mV the wave completes the complete oscillation: frequency **50Hz $\pm$ 10%***; if you increase the X scale (towards the second), the wave becomes like a festoon with scattered dots (appearance of a quantum field?), if vice versa you decrease the 180° which however do not cancel each other out but coexist, similar to the stylized appearance of a double helix.

These observations constitute a first approach to the detection of the intrinsic nature of the Quantum Void or Space, and its enormous energy content, preparatory to obtaining concrete results in the fields: energy, health, transport, communication, etc. where it is necessary to have the strength, courage and desire to invest in the creation of a research project dedicated to the in-depth analysis of this initial demonstrative research.

**Schumann resonances: the Earth and ionosphere are negative and positive parts, respectively, of a giant capacitor capable of generating very low electromagnetic frequencies. These resonances are excited by electrical discharges from lightning and were predicted mathematically by W. O. Schumann in 1952. Scumann's base frequency is 7.83 Hz, which, however, as a result of detections made by the Russian Space Observing System was calculated in 2014 to be 8.5 Hz. Multiplying 8.5 by 6 gives 51 (50Hz $\pm$ 10%)*

11) Spin-off from the Science of Resonances

The "Science of Resonances", as well as bringing an evolution in the knowledge of the various "whys" present in the "intimate nature of the Universe", will be preparatory towards a paradigm shift in various fields, such as:

- 1) Manifestation of a clean energy flow from Space
- 2) Improved health in biological life
- 3) Innovative Means of Transport
- 4) New topological materials, and new composite nano-materials.
- 5) Transmission of information

1) In the energy field, the very concept of Energy will change, which is not produced but manifests itself, through the mechanism of communicating vessels, see: **Ivaldi, Smigliani, 2024c**.

The "Science of Resonances" leads us to understand that our power plants do not directly supply energy to our systems, but they only serve to open a "door" and keep it open to allow free and clean energy to flow. present in the "Quantum Void" or "Space", which is conveyed by the electric generator through the cables to our various users, which then disperse it with their functioning back into the Space from where it comes. There is therefore a communication route between the power generator of the power station and our end user, which ensures that as soon as our user finishes its work, it sends a signal that immediately closes the "*open door*" towards Space; note that energy teleportation had already been investigated by Masahiro Hotta, theoretical physicist and assistant professor at Tohoko University in Japan, see: **Hotta, 2010**.

2) For the improvement of health in biological life, we note that in the "Science of Resonances" the "Quantum Vacuum" or "Space" is seen as a Living and Conscious entity, so by applying the vision inherent in the "Science of Resonances" to In our modern times, where technology and artificial intelligence are gradually replacing human work, using the digital reality of algorithms (which, in medicine, are called "Protocols"), the step forward will be notable, essentially for two reasons :

A) For the possibility of better diagnosing any dissonances and/or destructive interferences characteristic of pathologies (through the detection of non-normal frequencies and amplitudes, according to the Algebra of Nature).

B) In the absence of obvious structural deficiencies (complete fractures, amputations, etc.), the possibility of definitively treating them by restoring the correct resonances. This is evident especially if we think about DNA: the current interpretation of the genome it contains is oriented towards a list of characters and notes, thinking that, by replacing one of them, it can be engineered to obtain different answers (cyborgs, hybrid men, diseases avoided, etc: going forward at this rate all diseases will become "*genetic*", until we begin to treat only the perfectly healthy).

In reality, no one yet knows how to control any engineered genetic mutation, for the simple reason that there is no need to change anything. The "Science of Resonances" teaches that DNA is a piano which, if forgotten, can generate incorrect information that leads to certain diseases. In this case it is necessary to "*tune*" the piano to the absolute resonance of the Universe that the anomaly disappears. What could changing one or more keys on the piano, by people who are still

inexperienced in the vision of the "Science of Resonances", entail? The new sound that will be generated, what total mutation could it lead to?

Questions that will be able to find a reliable answer only when profound knowledge in the "Science of Resonances" is better developed.

3) Innovative Means of Transport, and in particular new means to move ourselves in Space, in which Space itself represents the "*fuel*", to move in a non-empty Quantum Void, as Einstein's Relativity would have it, which by replacing the depth of Space with Time, it eliminated the need for the presence of the Aether. Therefore, only by eliminating the Space-Time of Relativity, and bringing the Superfluid Void back into play as Ether, will it be possible to arrive at the true origin of the phenomena of interaction between electromagnetism and gravity, see: **Scala, 2022b**, where it is possible to originate from the Quantum Void phenomena of difference in ethereal density, see: **Fedi, 2017**.

The correlation between electromagnetism and gravity by means of the differential form of the gravitational field equations makes the production of negative gravitational fields possible in theory, and therefore usable for air and space propulsion, see: Page 8 of **Scala, 2007**, the studies of Doctor **Puthoff 2012**, and **Tajmar's studies, 2010**.

The "Science of Resonances" will be able to lead us towards the development of "*Innovative Means of Transport*", operating on an "*electromagnetic loop*" whose "*engine*" will be represented by their external body, which will be composed of two layers, of which the internal part will have the shielding function for passengers, which in current technology we can for example find in the characteristics of Kevlar and Teflon; while the external part, made up of an "*Energetic-resonant*" material, will have the function of generating and maintaining the "*Electromagnetic Loop*" or "*Energy Bubble*".

Inside the Innovative Means of Transport will be a trigger and control device via feedback on the energy level of the "*electromagnetic loop*", and a system for managing movements in space, which will work through the creation of an "*energy arrow*" with respect to the "*center of energy balance*" of the mass of the Innovative Means of Transport, which, in order to "*rebalance*" itself in Space, will generate the movements of the Means in the pre-established direction.

Inside the Innovative Means of Transport, there will be great comfort for passengers, given that in the "*Energy Bubble*" an area is created that is energetically independent from the external space, which will not cause the effects generated by the inertia of travel to be felt.

4) New topological materials, of which the best known is Graphene, see: **Rayhan, Begum, 2024**; and new composite nano-materials, which find an interesting example in the history of Microclusters see: **Flanagan, 2000**, which present a new state of matter, in which atoms are not seen as "*particles*" suspended in an empty space, but they are seen as vortices of the aether.

The story of "Microclusters" begins with the article in the December 1989 edition of Scientific American, which summarily said: Divide and subdivide a solid, and the characteristic features of its solidity will disappear one by one like the somatic features of the "*Cheshire Cat* ", to be replaced by characteristics that are no longer those of liquids or gases, but belong to a new state of matter, and pose questions to the physics and chemistry of the solid state, such as:

A) How small must an aggregate of particles become before the character of the original substance is lost?

B) How can atoms reconfigure themselves if freed from the influence of the matter that surrounds them?

C) If the substance is a metal, how small must a group (=cluster) of atoms be to lose the characteristic sharing of free electrons that underlies conductivity?

Less than two years after this story broke into the world of official science, the physics of Microclusters was perfected by Sugano, 1991, where it was explained that we can assemble groupings of atoms into four basic size categories, each of which with different property:

Molecules: 1-10 atoms,

Microcluster: 10-1000 atoms,

Fine particles: 1000-100,000 atoms,

Mass: more than 100,000 atoms.

From observing this list we would expect that Microclusters have traits in common both with molecules and with the fine structure of matter; instead they show us properties that neither one nor the other possess; since Microclusters composed of 10 to 10³ atoms exhibit neither the properties of the corresponding mass, nor those of the corresponding molecules of a few atoms.

It can therefore be said that Microclusters form a new state of matter, which is a cross between microscopic solids and microscopic particles such as atoms and molecules, and present anomalous properties such as:

A) They burn at a much higher temperature than elements of our traditional chemistry which are consumed in a maximum of 70 seconds, while Microclusters must be burned for more than 200 seconds to appear an analyzable color spectrum.

B) When, through a technological process, we obtain a fragment of "Microcluster" with the desired shape and size, and with a radius of the order of 10 angstroms; since its atoms and molecules are very stable, we note that its properties can be measured; while the same kind of measurements appear to be impossible for fine particles existing in nature.

Currently, laser technology is used to obtain Microclusters, but in the future technologies linked to the "Science of Resonances" will certainly be used, and these processes will be the same ones that will be used to create future "*Resonant with Space*" materials, such as it is Graphene, which has peculiarities recognized in the international scientific community; in particular, graphene powder is capable of forming three- and two-dimensional aggregates, therefore with SP³ and SP² hybridization.

The latter allows electrons to slide on the graphene, where energy gaps form, in a two-dimensional form, such as the flat bands also highlighted in a study by MIT in Boston, where they managed to trap two electrons in a pyrochlore crystal, thus obtaining the equivalence of their two energy states in an electronic flat band.

Again from these studies it emerged that the movement of the carbon-carbon sigma (σ) bond axis by at least 1-3 degrees induced, in the graphene layer, the formation of energy gaps capable of making a flow of energy flow with minimal resistance. electrons, as described in Chapter 11).

5) For Sciences and Technologies in the sector of information transmission, we see that today to transmit through the ether we use electromagnetic waves, which can create negative consequences towards our harmonic energetic balance, including the decrease in our immune defenses.

A less invasive system towards our biological being could be the one that uses the "*Gravitational Tensors*" hypothesized in the symmetrized Maxwell equations of Eng. Sabato Scala, see: **Scala, 2022b**, where the electromagnetic wave is always accompanied by the gravitational component, which expands according to the geometric shape of a logarithmic spiral, and where through the fifth dimension, all the universal information that occurs in real time (Entanglement), therefore much faster than the manifestation of electromagnetic waves which travel at the speed of light.

The "*Gravitational Tensor*" (yet to be discovered) could be composed of a set of space quanta as large as the Planck length, which generate a sort of inextensible thread that can extend infinitely; so when a voltage is applied at any point of the tensor, instantaneously this voltage can appear regardless of the distance at any point of the tensor.

This hypothesis is also confirmed in the observations on the behavior of dark matter, observed in the distribution of the filaments of galaxies, where the central part, unlike what was expected, is not faster than the external one, and the internal parts seem held together by filaments that give rise to the famous spiral tails clearly visible to all, see: **Khoury, 2022**.

The "*Gravitational Tensors*" are also those on which matter in the universe is organized according to very specific paths, which are also reflected in the De Broglie - Bohm pilot wave theory.

11) Conclusions

If we look at modern science as a whole, we observe that it still functions as Aristotle structured it, in an axiomatic and deductive way, where the axioms are not demonstrable, they are assumed only because they are generally considered evident; from which it can be deduced that truth and demonstrability are not the same thing.

In fact, Gödel's theorem demonstrates that semantic truth depends on how the meaning is defined, which is negotiable; from which it can be deduced that: "*By definition, absolute truth cannot exist within our passage into imperfection; while it exists in the sixth dimension where the Perfect One resides.*"

Therefore observation and experimentation can set a limit to the admissible scientific hypotheses, but they alone are not able to determine the limits of science; where for classical science the concept of law is a discipline aimed at understanding and representing the functioning of Nature, in which through the examination of experiments, it classifies its results, and then transforms them into absolute laws. While the "Science of Resonances" teaches us that to understand the mechanisms inherent in the "intimate nature of the Universe", we need the interpretation of the various "*whys*", which lead us to discover that the Universe is an enormous mind in perfect equilibrium on itself, which follows an order like what **Bohm, 2006**, called "*implicated order*", in which to represent the evolutionary functioning of the intimate nature of the Universe, the presence of a form of intrinsic Universal Intelligence is needed, see: **Ivaldi 2021, Meijer and Ivaldi 2022, Geesink and Ivaldi 2023**, where the concept of "*concrete particle*" is replaced with that of

"field and resonance", so if we observe an atom, a flower or a galaxy, we will always and only find ourselves inside geometric shapes in continuous evolutionary mutation.

Finally, the "Science of Resonances" will lead us to better understand and internalize what this Universe is, and consequently also what we are, and what we are doing in this wonderful and harmonious Brilliant Garland called the Universe, in which we all men and women nature we are linked by a common thread, and where the spin-offs deriving from the "Science of Resonances", inserted into the fabric of our society, will produce the effect of a "*beneficial Virus*", which by spreading will mark the birth and evolution of "*The era of Good Living*".

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